

**LOMBA KOMPETENSI SISWA (LKS)
SEKOLAH MENENGAH KEJURUAN
TINGKAT WILAYAH JAKARTA TIMUR 2 TAHUN 2025**

MODUL C

**INFRASTRUCTURE PROGRAMMABILITY &
AUTOMATION**



**Bidang Lomba
IT NETWORK SYSTEMS ADMINISTRATION**

Introduction

Accurate and up-to-date documentation has always been a challenge in IT. With multiple engineers working on the same systems, it is hard to keep track of who changed what. Applix Corporation decided to fix this problem and hired you to modernize, harden, and extend their infrastructure.

Description of Project and Tasks

You will be migrating VMs to Infrastructure as Code (IaC) and simplifying the process of creating new services. You have access to development VMs (DEV-LIN). These VMs can be used for developing and testing your work.

Login for all VMs and Devices:

Linux

Username : root/user

Password : P@ssw0rd

Windows

Username : administrator/user

Password : P@ssw0rd

Instructions to the Competitor

Part 0. Basic Configuration

Install and configure all VMs according to the appendix.

Part 1. Linux

Use Ansible to configure the Linux hosts LIN[1-2] from **BASTION VM**.

Create file `/home/user/workfolder/hosts` as the ansible inventory. For marking, all playbooks will be run in order using the command `"ansible-playbook -i linux/playbookname.yml"` in the `/home/user/workfolder` directory as user **user**.

Variables like "hostname" and "webport" in `/home/user/workfolder/hosts` are subject to change for marking.

General

- Create a directory `/home/user/workfolder/linux`
 - Place all playbooks directly inside this directory.
 - You are free to create folders/files in this directory for running the playbooks.
 - When you run the playbooks for the second time, every task should either be in the "ok" state or be "skipped."

Common Role

- Create an ansible role called "common" for installing common packages on linux.
 - The common packages are: curl, htop, iotop, dnsutils and net-tools

Basic Configuration

- Create a playbook called **1-basic-configuration.yml** for configuring the basic configuration
 - Set hostname based on "hostname" variable that defined on the inventory
 - Include common role to this playbook and add tags: common to independently run the role

Custom MOTD (Message of the Day)

- Create a playbook called **2-custom-motd.yml** for updating MOTD message
 - Use jinja2 templating to create custom MOTD message
 - MOTD file should be located in `/etc/motd`
 - MOTD message must be dynamic, here is the content:

```
Welcome to {{ hostname }} server!

Please be careful when making any changes!

Support: {{ support_contact }}
```

- Secure the `support_contact` variable using Ansible Vault and render it in the MOTD jinja2 template without exposing it in plaintext in the playbook or inventory.
 - `support_contact` containing support number **08123456789**
 - use **P@ssw0rd** as an ansible vault pass.

Web Server

- Create a playbook called **3-web-server.yml** for configuring multiple web servers (**nginx**)
- Install a web service on all hosts in the group “web”
 - The local website should listen to the port “webport” variable in /home/user/workfolder/hosts and can be accessed through their IPs.
 - Display the following content

“Hello from <hostname> !”

Users

- Create a playbook called **4-users.yml** for importing users on all LIN hosts
 - Create user-xxx where xxx is 001-999. Password should be the same as the username
 - If the user number is **even**, assign **/bin/bash** as the login shell. (e.g. user-002 will get /bin/bash as shell)
 - If the user number is **odd**, assign **/bin/sh** as the login shell. (e.g user-001 will get /bin/sh as shell)
 - Make sure, that the password is not changed if there is already an existing user with the same username and UID

Part 2. Windows

Use Ansible for configuring the Windows hosts, labeled as WIN, from the BASTION VM.

Create file **/home/user/workfolder/hosts** as the ansible inventory. For evaluation purposes, all playbooks will be executed in sequence using the command “ansible-playbook -i hosts windows/playbookname.yml” within the /home/user/workfolder directory as user **user**.

General

- Create a directory **/home/user/workfolder/windows**
 - All playbooks should be located at the root of this directory
 - You are free to create folders/files in this directory for running the playbooks
 - When you run the playbooks for the second time, every task should either be in the “ok” state or be “skipped.”

Hostname

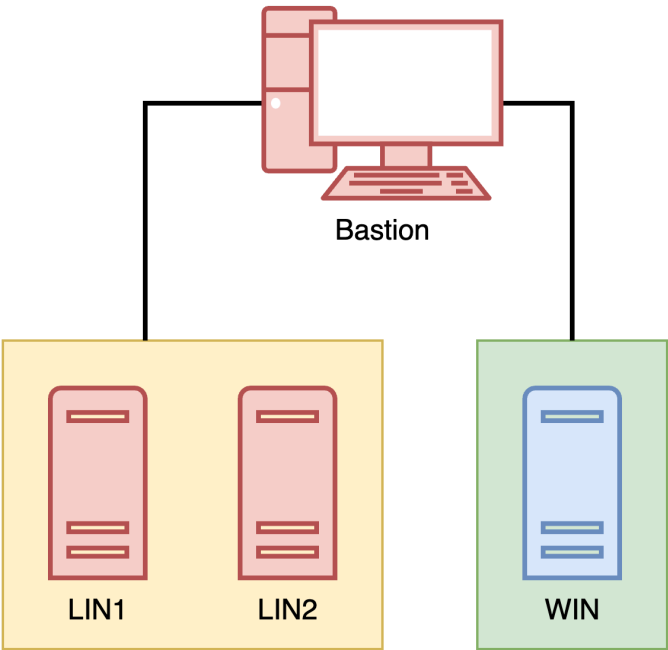
- Create a playbook called **1-hostname.yml** for configuring the hostname
 - All hosts should receive the hostname based on the “hostname” variable in /home/user/workfolder/hosts file

Security and logging

- Create a playbook called **2-sec-log.yml** for configuring security settings
 - Stop and disable the RDP Service on all Windows hosts

Appendix

Topology



System table

ID	IP	OS	DESCRIPTION
BASTION	10.0.10.1	Debian 12.9 (GUI)	Main machine to run ansible playbook
LIN1	10.0.10.11	Debian 12.9 (CLI)	Dynamic Config
LIN2	10.0.10.12	Debian 12.9 (CLI)	Dynamic Config
WIN	10.0.10.20	Windows Server 2022 (Desktop)	Dynamic Config